

Mathematical Analysis II

Diffeomorphisms, Inverse & Implicit Function Theorems

Yuewen Chen

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1 Coordinate Transformations and Diffeomorphisms

To study local invertibility, we first establish the language of coordinate transformations and their differentials.

Example: Polar Coordinates

Consider the standard polar transformation $(r, \theta) \mapsto (x, y) = (r \cos \theta, r \sin \theta)$. Its Jacobian matrix and determinant are given by:

$$J\Phi(r, \theta) = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad \det J\Phi(r, \theta) = r > 0$$

Since the determinant is non-zero (for $r > 0$), the differential is invertible. By the chain rule, the inverse Jacobian is explicitly:

$$J(\Phi^{-1})(x, y) = (J\Phi)^{-1} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\frac{1}{r} \sin \theta & \frac{1}{r} \cos \theta \end{pmatrix}$$

Definition: Diffeomorphism

A bijective mapping $\Phi : \Omega \rightarrow \Omega'$ is called a C^m -diffeomorphism ($m \geq 1$) if both the mapping and its inverse are of class C^m .

$$\Phi \in C^m(\Omega, \Omega') \quad \text{and} \quad \Phi^{-1} \in C^m(\Omega', \Omega)$$

Invariance of the Differential:

For any scalar function f , its differential remains invariant under arbitrary coordinate transformations (pullback $\Phi^* f$):

$$df = \sum_{k=1}^n \frac{\partial f}{\partial y_k} dy_k = \sum_{j=1}^n \frac{\partial f}{\partial x_j} dx_j$$

2 Inverse Function Theorem: Modified Newton's Method

The following strictly establishes the Inverse Function Theorem by directly employing the Modified Newton's Method at an arbitrary base point without coordinate translation.

Theorem 7.5.2 (IFT - Restated):

Let $f \in C^1(\Omega, \mathbb{R}^n)$ and $p = x_0 \in \Omega$. If the Jacobian matrix $A := df(x_0)$ is invertible, then there exist open neighborhoods U of x_0 and V of $y_0 = f(x_0)$ such that the restriction is a diffeomorphism:

$$f|_U : U \rightarrow V \quad \text{is a diffeomorphism.}$$

The Operator Formulation (Freezing the Jacobian):

To solve $f(x) = y$, we define the iterative scheme by freezing the Jacobian matrix at the initial point x_0 . We rewrite the scheme as a fixed-point problem $x = T_y(x)$:

$$T_y(x) := x - A^{-1}(f(x) - y)$$

Observe the fundamental equivalence: $T_y(x) = x \iff f(x) = y$.

Contraction Factor q & Invariant Ball:

Since $f \in C^1$, the differential $df(x)$ is continuous at x_0 . There exists a radius $r > 0$ such that for all $x \in \overline{B}(x_0, r)$, the operator derivative satisfies:

$$\|dT_y(x)\| \leq \|A^{-1}\| \cdot \|A - df(x)\| \leq \frac{1}{2} := q$$

Let $\delta = \frac{r}{2\|A^{-1}\|}$. If the target y is bounded by $\|y - y_0\| \leq \delta$, the Mean Value Inequality ensures T_y maps the closed ball into itself:

$$\|T_y(x) - x_0\| \leq q\|x - x_0\| + \|A^{-1}(y - y_0)\| \leq \frac{1}{2}r + \frac{1}{2}r = r$$

Convergence & A Priori Error Estimate:

By the Contraction Mapping Theorem, the sequence $\{x_n\}$ converges linearly to the unique fixed point x^* . The standard contraction result yields the rigorous a priori bound:

$$\|x_n - x^*\| \leq \frac{q^n}{1 - q} \|x_1 - x_0\| = 2^{1-n} \|A^{-1}(y - y_0)\|$$

Regularity Lemma & Differentiability of the Inverse:

The contraction mapping guarantees the existence and continuity of the local inverse f^{-1} . To upgrade it to a differentiable mapping, we invoke the Regularity Lemma:

If f is a homeomorphism, $f \in C^1$ and $df(x)$ is invertible $\implies f^{-1} \in C^1$

Since $\|I - A^{-1}df(x)\| \leq 1/2 < 1$ inside our invariant ball, $df(x)$ is guaranteed to be non-singular for all $x \in U$. Thus, the local inverse is C^1 , completing the proof that $f|_U$ is a C^1 -diffeomorphism.

Higher-Order Smoothness (C^m Inverse Function Theorem):

This result naturally generalizes to higher degrees of smoothness. For any $m \geq 1$:

$$f \in C^m(\Omega, \mathbb{R}^n) \text{ and } \det df(p) \neq 0 \implies f^{-1} \in C^m(V, U)$$

This is established by the explicit differentiation of the identity $f(f^{-1}(y)) = y$:

$$d(f^{-1})(y) = [df(f^{-1}(y))]^{-1}$$

Since matrix inversion is a smooth operation (derived from Cramer's Rule as rational functions of determinants), mathematical induction on m guarantees that f^{-1} shares the exact same smoothness class as f .

3 True Newton's Method and Quadratic Convergence

While the Modified Newton's method guarantees existence (requiring only C^1), practical evaluation of the inverse requires the True Newton's method, which updates the Jacobian at each step and requires C^2 smoothness.

The True Newton Iteration:

Assuming $F \in C^2(\Omega, \mathbb{R}^n)$, the iteration to solve $F(x) = b$ is given by:

$$x_{k+1} = x_k - [JF(x_k)]^{-1}(F(x_k) - b)$$

Quadratic Error Evolution (Taylor Remainder):

Let x^* be the true root and define the error $e_k = x_k - x^*$. Expanding $F(x^*)$ around x_k with a second-order Taylor remainder R_1 :

$$0 = F(x^*) = F(x_k) + JF(x_k)(x^* - x_k) + R_1(x_k)$$

Since $F \in C^2$, there exists $M > 0$ such that $\|R_1(x_k)\| \leq M\|e_k\|^2$. Substituting this back into the Newton step perfectly cancels the first-order terms:

$$e_{k+1} = x_{k+1} - x^* = [JF(x_k)]^{-1}R_1(x_k)$$

Safety Zone and Convergence Bound:

By the Perturbation Lemma (Neumann Series), the inverse Jacobian norm is uniformly bounded by some $\beta > 0$ near x^* . Taking norms yields the quadratic convergence rate:

$$\|e_{k+1}\| \leq \beta\|R_1(x_k)\| \leq (\beta M)\|e_k\|^2 := C\|e_k\|^2$$

If the initial guess x_0 satisfies $C\|e_0\| < 1$, mathematical induction guarantees the sequence never leaves the invertible safety zone, collapsing the error geometrically.

4 The Implicit Function Theorem

The Implicit Function Theorem can be elegantly established as a direct corollary of the Inverse Function Theorem by lifting the dimension space using an augmented mapping.

Theorem Statement:

Let $F : \Omega \subset \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a C^k mapping ($k \geq 1$) near (x_0, y_0) . Assume $F(x_0, y_0) = 0$ and the partial Jacobian matrix with respect to y is invertible:

$$\det J_y F(x_0, y_0) \neq 0$$

Then $F(x, y) = 0$ uniquely determines a C^k implicit function $y = f(x)$ near x_0 .

Proof via Augmented Mapping (Block Matrix Technique):

Construct a new mapping $G : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^{n+m}$ that artificially perfectly preserves the x coordinates:

$$G(x, y) := (x, F(x, y))$$

The full Jacobian matrix of G at (x_0, y_0) is a block upper-triangular matrix:

$$JG(x_0, y_0) = \begin{pmatrix} I_n & 0 \\ J_x F & J_y F \end{pmatrix}$$

Reduction to Inverse Function Theorem:

The determinant of this block matrix is exactly the determinant of the sub-block $J_y F$:

$$\det JG = \det(I_n) \cdot \det(J_y F) \neq 0$$

Since JG is invertible, the Inverse Function Theorem guarantees that G has a local C^k inverse $G^{-1}(u, v) = (u, \gamma(u, v))$. Setting $v = 0$ precisely extracts the implicit function:

$$(x, y) = G^{-1}(x, 0) \implies y = \gamma(x, 0) := f(x)$$

Derivative of the Implicit Function:

Differentiating the identity $F(x, f(x)) \equiv 0$ via the multivariable Chain Rule yields the explicit differential formula:

$$Jf(x) = - \left[J_y F(x, f(x)) \right]^{-1} J_x F(x, f(x))$$